Avinash Maskey

Lecturer in Information Technology

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Research Interests

Digital platforms and fintech. I am interested in how people and firms adopt and use digital payments, mobile wallets and online trading platforms, and how they react when these systems fail or when regulations change suddenly.

Causal inference and machine learning for economic data. I want to know whether a model's predictions actually lead to better decisions, not only whether they are accurate. In my research on the Nepal Stock Exchange (NEPSE), forecasts that looked accurate gave no real benefit once trading fees were included. This is why I use methods such as Double Machine Learning and compare models with simple benchmarks.

Education

Master of Computer Information System (MCIS) Mar 2020 – Feb 2024

Nepal College of Information Technology, Pokhara University Nepal

- CGPA 3.88/4.00, batch topper; university topper in the third semester (4.00/4.00).
- Thesis: *Forecasting Stock Exchange Sector Indices Using Granger Causality and Vector Autoregressive Models* (supervisor: Dr. Suresh Pokharel). An adaptive VAR with an empirically chosen window outperformed static and fixed rolling-window models.

Bachelor of Information Management (BIM) Oct 2014 – Oct 2018

Kist College of Management, Tribhuvan University Nepal

- CGPA 3.61/4.00; university top five in the fifth and eighth semesters; college semester topper in the fifth, sixth and eighth semesters.

Research

WORKING PAPERS AND WORK IN PROGRESS

[1] **Maskey, A.** (2026). Rising connectedness, weak lead-lag: Evidence from Nepal Stock Exchange sector indices using econometric and machine learning methods. *Working paper.* [Paper page]

- Built a Python pipeline with automated tests that reproduces every result from frozen raw data on ten sector indices (2018–2026). An audit against 3,844 trading days showed that one- and two-day date shifts in a compiled dataset were enough to produce spurious Granger causality.
- Compared random forests and XGBoost with random walk, AR and VAR benchmarks over a 500-day out-of-sample period. The tree models predicted direction on up to 62% of days, and SHAP values identified the inputs behind their forecasts.
- Assessed the forecasts by the investment decisions they would support. After NEPSE transaction costs, none of the direction strategies outperformed buy and hold.
- Measured Diebold–Yilmaz connectedness with rolling and time-varying parameter VARs. Total connectedness rose from 26.5% in 2016 to 77.7% in 2022.

[2] **Maskey, A.** (in progress). Metacognitive learning behaviors in LMS-supported learning environments: A Delphi study and questionnaire development.

- Conducting a Delphi study with subject experts and developing a questionnaire based on it.

PEER-REVIEWED JOURNAL ARTICLES

[3] **Maskey, A.** (2026). Resilience of Nepal's stock market to systemic shocks: A case study of COVID-19. *SEBON Journal*, 12(1), 59–92. [Full text]

- Estimated the effect of the pandemic period on daily NEPSE returns using Double Machine Learning with Ridge regression and five-fold cross-fitting on 1,371 trading days (2018–2023), controlling for macroeconomic, policy and market variables.
- Daily returns were about 21 basis points higher than the macroeconomic counterfactual. The result held under alternative specifications, GARCH models, event studies, sensitivity analysis and 1,000 placebo permutations.

[4] Shrestha, N., Manandhar, N. K., Bhandari, B., **Maskey, A.**, & Giri, P. R. (2026). Undergraduate students' perceptions of generative artificial intelligence (Gen AI) in academic assignments. *Advances in Mobile Learning Educational Research*, 6(1), 1809–1823. doi:10.25082/AMLER.2026.01.014

CONFERENCE PRESENTATION

[5] **Maskey, A.**, & Khatiwada, G. (2026). Stock market awareness and participation among university students: A case of Kathmandu University. *1st Joint National Conference on Capital Market in the AI Era*, Nepal.

- Analysed survey data from 153 students with logistic and linear regression. Students with high awareness were about six times more likely to participate in the market.

Academic Appointments

Lecturer (full-time) Jul 2025 – present
Department of STEAM Education, School of Education, Kathmandu University Kathmandu, Nepal

- Teach Artificial Intelligence, Management Information System, Database System, System Analysis and Design, and Principles of Internet Technologies and Web Applications in the BTech Ed (IT) and BIT programs, and AI in STEAM Education in the Master of STEAM Education.
- Supervise undergraduate research projects in machine learning, educational technology and information systems, as well as capstone projects and internships.
- As curriculum designer since October 2024, developed eight syllabi, including AI in STEAM Education (master's level), Web Engineering, E-Governance and Professional IT Project II.
- Serve as Focal Person for the BTech Ed (IT) programs at five partner institutions.

Visiting and Adjunct Faculty (part-time) Nov 2019 – Jul 2025
Kathmandu University and colleges affiliated with Tribhuvan and Purbanchal Universities Kathmandu, Nepal

- Taught database systems, software engineering, web technology, system analysis and design, e-commerce, e-governance, MIS and ethics in IT at eight institutions, and supervised internships and undergraduate projects.

Industry Experience

Co-founder; Project Manager, later Director of Product Management Jan 2022 – present
Prajanya Soft Pvt. Ltd. Kathmandu, Nepal

- Worked full-time until July 2025 and continue as co-founder. Responsible for product decisions on web and mobile systems in fintech, healthcare and e-commerce for more than ten clients in Nepal and abroad.
- Managed the development of a hospital management ERP (patient records, diagnostics, billing and inventory) and an e-commerce platform with inventory and order workflows.
- Coordinated requirements and architecture for an Australia-based trade services platform with distributed teams, and planned priorities and releases with engineering, design and QA staff.

Financial Analyst, Trading and Research Department Mar 2019 – Jul 2019
Kriti Capital & Investments Ltd. (now Machhapuchchhre Capital) Kathmandu, Nepal

- Analysed macroeconomic and development bank data, and prepared fundamental and technical analyses of listed stocks for the trading team.

IT Intern, IT Department Apr 2018 – Aug 2018
Century Commercial Bank Ltd. (now Prabhu Bank) Kathmandu, Nepal

Methods and Technical Skills

Causal inference	Double Machine Learning with cross-fitting, event studies, placebo and sensitivity analysis
Time series	VAR, Granger causality, impulse responses, Diebold–Yilmaz connectedness, TVP-VAR, GARCH, ARIMA
Machine learning	Random forests, gradient boosting (XGBoost), SHAP, out-of-sample evaluation with Diebold–Mariano tests
Survey research	Scale reliability, t-tests, ANOVA, logistic and linear regression, Delphi method
Programming	Python (pandas, NumPy, scikit-learn, statsmodels, XGBoost, SHAP), R, SQL, Java, JavaScript, PHP; Git; automated tests for data pipelines
Languages	English (fluent; English-medium education from school through master's degree), Nepali (native), Hindi (fluent)

Honors and Leadership

- Batch Topper, Master of Computer Information System, NCIT, Pokhara University (2024).
- President of the IT Club (2017–18) and the Sports Club (2016–17), Kist College; member of the organizing committee of a national-level workshop on machine learning and data science (2018).

References Available On Request.